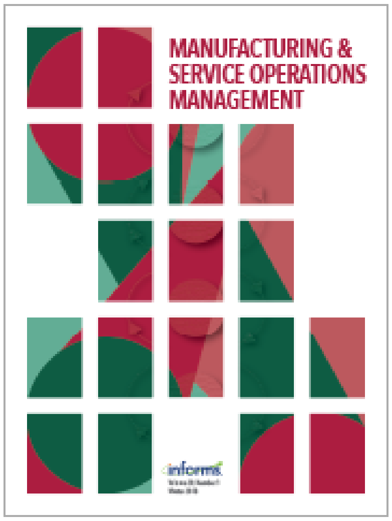

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Improving Farmers' Income on Online Agri-Platforms: Evidence from the Field

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Abstract. ***Problem definition*:** To improve the welfare of smallholder farmers, multiple countries (e.g., Ethiopia and India) have launched online agri-platforms to transform traditional markets. However, there is still mixed evidence regarding the impact of these platforms and, more generally, how they can be leveraged to enable more efficient agricultural supply chains and markets. This paper describes work conducted in close collaboration with the state government of Karnataka, India, to *design*, *implement*, and *assess* the impact of a new two-stage auction on the state's online agri-platform, the United Market Platform (UMP). ***Methodology/results*:** To ensure implementability and protect farmers' revenue, the auction design is guided by practical operational considerations, as well as semistructured interviews with a majority of the traders in the field. A new behavioral auction model informed by the field insights is developed to determine when the proposed two-stage auction can generate a higher revenue for farmers than the traditional single-stage, first-price, sealed-bid auction. The new auction mechanism was implemented on the UMP for a major market of lentils in February 2019. By the end of May 2019, commodities worth more than $6 million (USD) had been traded under the new auction. A difference-in-differences analysis demonstrates that the implementation has yielded a significant 3.6% price increase (corresponding to a 55%–94% profit gain) for over 10,000 farmers who traded in the treatment market. ***Managerial implications*:** The results from this paper offer tangible lessons on how innovative price discovery mechanisms could be enabled by online agri-platforms in resource-constrained environments. Importantly, the success of these designs critically depends on careful considerations of systemic operational and behavioral factors that affect trades in the physical markets.

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Keywords: poverty alleviation • smallholder farmers • two-stage auction • field implementation • developing countries • difference-in-differences • multimethod • behavioral operations • anticipated regret • level-*k* thinking • sustainable operations

1. Introduction

The phenomenon of severe poverty continues to persist among smallholder farmers in developing countries, in part due to unfavorable market outcomes for these farmers (Mondiale 2008). Prior studies suggest that imperfect competition and inefficient price discovery processes in traditional agricultural markets are important factors affecting farmers' income (Goyal 2010, Bergquist and Dinerstein 2020). To tackle these challenges, one prevalent intervention that has been attracting substantial investment is to connect geographically isolated markets via an online platform. In such a platform, various aspects of the price discovery process are digitized and automated, including winner determination and declaration in auctions, as well as dissemination of price information. The hope is that these platforms could increase market competition among buyers, enable transparency of the price discovery process, and, ultimately, improve farmers' income.

Multiple countries have launched such online agri-platforms—for example, the commodity exchange platforms in Ethiopia, Kenya, Nairobi, and Uganda; the *e*National Agriculture Market (eNAM) launched by the central government of India; and the Unified Market Platform (UMP) implemented in the state of Karnataka, India. The price discovery mechanisms used in these platforms vary widely, including warehouse-based negotiation, ascending auctions, and first-price sealed-bid auctions, mostly following a digital version of the traditional (preplatform) mechanisms. It remains an open question as to which mechanism may be most beneficial for farmers (Strzebicki 2015, Deichmann et al. 2016). This is further highlighted by the mixed empirical evidence regarding the impacts of these platforms on farmers' income. For example, trades on the Ethiopia Commodity Exchange are dominated by two export crops, coffee and sesame seeds, whereas farmers of other crops are largely left out (*Economist* 2017, Hernandez et al. 2017). In India, only 14% of all farmers have registered on eNAM, and over half of these registered farmers are reported to not have benefited from the platform (Business Line 2019).

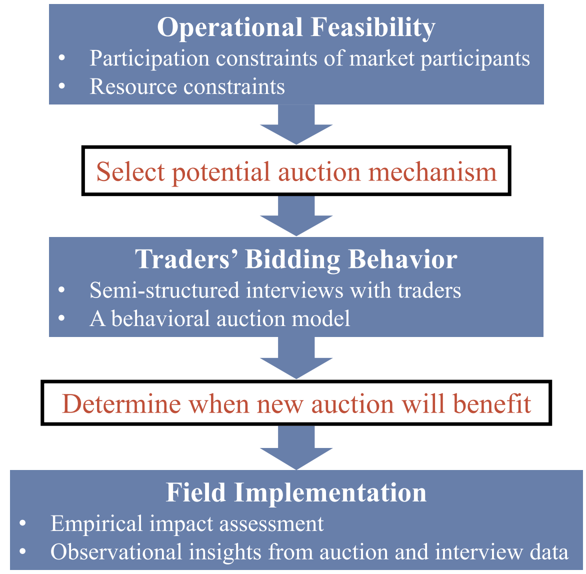
This paper describes work done in close collaboration with the state government of Karnataka to increase farmers' revenue by redesigning the price discovery mechanism on the UMP. The UMP was created in 2014 by the state government to unify all trades in agricultural wholesale markets in Karnataka with a single digital platform. Farmers sell their products in these markets to traders through first-price sealed-bid auctions conducted on the UMP. By November 2019, approximately 62.8 million metric tons of products valued at $21.7 billion (USD) had been traded across 162 markets on the UMP. The major types of products traded on the UMP include grains, lentils, oil seeds, and cash crops (e.g., cotton, areca nut). Recent research analyzing the UMP shows that although the UMP has brought significant price benefits for farmers of certain products (groundnut, maize), farmers of other products (e.g., lentils) have not benefited from the platform. A potential reason for the lack of benefit is that these products are sold in geographically isolated markets, with low levels of competition among a small number of local traders in each market (Levi et al. 2020a). Therefore, we seek to utilize auction (re)design as a means to increase trader competition for these products and, consequently, improve farmers' revenue.

The literature on auction design is extensive and has analyzed "optimal" mechanisms with appealing theoretical properties. The well-known revenue equivalence theorem implies that the first-price sealed-bid auction would result in equivalent expected revenue to many other auction mechanisms when bidders satisfy certain rationality assumptions (Krishna 2009).[1] However, empirical evidence suggests otherwise: Human bidders are subject to behavioral regularities that substantially influence revenue outcomes (see Kagel and Levin 2016 and Elmaghraby and Katok 2018 for recent reviews). Because our goal is to use auction design to materially increase farmers' revenue, it is imperative to investigate and account for potentially salient behavioral factors in the design process. Furthermore, to ensure implementability, the design must also take into account the operational constraints in the specific market at hand. This is especially important for resource-constrained environments, such as agricultural markets in developing countries, because it is observed that simply replicating what might be successful in more affluent settings does not work well (Alemu and Meijerink 2010, Business Line 2019).

In this paper, we employ a multimethod approach to design, implement a new auction mechanism on the UMP, and empirically verify that it effectively increases farmers' revenue over the legacy first-price sealed-bid auction design on the platform (Figure 1). We begin by carefully considering operational constraints in the markets to determine what mechanisms are even feasible. In particular, the design must get buy-ins from all market participants, including both farmers and traders, and it should also account for the limited resources available for training and technical changes. This analysis leads to the proposal of a two-stage auction design. The next step is to determine when adopting the two-stage auction will indeed benefit the farmers. To do so, we carefully examine the traders' bidding behavior based on semi-structured field interviews and the analysis of a resulting behavioral auction model. The analysis shows that the two-stage auction would most likely benefit farmers for products of constrained supply or high economic values. Eventually, tur (pigeon pea) is chosen as the product to implement the two-stage auction, due to its limited supply relative to its high demand across India.

Working with the state government of Karnataka, the two-stage auction was implemented on the UMP in February 2019 for a major market of tur. By the end of May 2019, more than 8,000 metric tons of tur worth over $6 million (USD) had been traded under the new auction

Figure 1. (Color online) (Re)designing the Auction Mechanism on the UMP



in the treatment market. To evaluate the impact of the implementation on farmers' revenue, a difference-in-differences analysis is performed utilizing lot-level bidding data (a total of 172,342 bids) for all tur trades in the treatment market and a comparable control market. The empirical analysis demonstrates that the implementation has yielded a statistically significant, 3.6% average price increase in the treatment market. The price increase translates into an average revenue gain of $346,000 (USD) for over 10,000 farmers in a matter of three months, representing an average 11% increase in their monthly income. Given low profit margins for these farmers (4%–7%), the range of profit improvement is substantial (55%–94%). These positive outcomes are particularly significant and encouraging given the lack of a price gain for tur farmers from UMP's original implementation in 2014 (Levi et al. 2020a).

2. Related Literature and Contributions

This paper addresses the crucial topic of improving smallholder farmers' welfare and makes important contributions to both research and practice. Our work is related to three streams of literature. The first stream is a growing body of operations management (OM) research focusing on improving farmer and social welfare in agricultural supply chains (see Bouchery et al. 2017 and Kalkanci et al. 2019 for broader reviews). Although a majority of these studies focus on the production stage (e.g., Dawande et al. 2013, Alizamir et al. 2019, Boyabati et al. 2019, Federgruen et al. 2019, Hu et al. 2019, Levi et al. 2020b), a few recent papers examine the market side of agricultural supply chains (Parker et al. 2016; Liao et al. 2019; Levi et al. 2020a, 2022; Gupta and Bansal 2022). Closely related to this body of OM research, scholars in developmental economics have also studied digital solutions for agriculture (Klerkx et al. 2019, Suri and Udry 2022). They explore the potential benefits, challenges, and constraints of adopting Information and Communication Technologies (Aker 2011, Fabregas et al. 2019), with a focus on agricultural extension services (Fu and Akter 2016), mobile technology (Evans 2018), and innovation platforms (Schut et al. 2018). They offer insights into the factors influencing technology adoption (Evans 2018), as well as the impacts of digital agriculture on poverty reduction, gender dynamics, and the overall innovation process in the agricultural sector (Sunding and Zilberman 2001). Our paper adds to the above streams of literature by examining auction design in the new context of online agricultural platforms. We demonstrate the importance and value of adopting a behavior-centric, field-based, multimethod approach for practical mechanism design. Our research shows that auction design that carefully accounts for salient behavioral and operational factors in the market can yield significant societal benefits.

The second stream of related literature studies the effects of bidders' behavioral factors on their bidding strategies. Kagel and Levin (2016) and Elmaghraby and Katok (2018) provide excellent reviews. Most of the research here uses laboratory experiments to examine various behavioral factors, such as anticipated regret, anchoring, risk aversion, learning, level-*k* thinking, and competitive arousal (e.g., Maskin and Riley 1984, Ariely and Simonson 2003, Ku et al. 2006, Crawford and Iriberri 2007, Filiz-Ozbay and Ozbay 2007, Engelbrecht-Wiggans and Katok 2008, Adam et al. 2015). We add to this literature in two key aspects. First, we extend the investigation of behavioral factors in auctions from laboratory experiments to a field context with professionals. Our field research suggests that the behavioral factor of anticipated regret widely observed in laboratory experiments of single-stage auctions also influences agricultural traders in two-stage auctions in the field. Agricultural traders may experience regretful emotions if they realized ex post that they could have had a chance to win the auction with a positive payoff, and anticipating such regretful emotions can stimulate more competitive bidding in the first place. Second, we leverage the field insights to develop a new behavioral auction model that both captures anticipated regret and applies the nonequilibrium framework of "level-*k*" thinking to analyze the traders' bidding strategies. Extensive experimental evidence shows that level-*k* thinking can better predict human behavior than equilibrium models do in various strategic contexts, including auctions (Camerer et al. 2004, Crawford and Iriberri 2007, Crawford et al. 2013). The fundamental idea is that people do not perform infinite iterations of best-response reasoning; instead, they are limited in their ability to anticipate others' strategic behavior. To the best of our knowledge, this paper is the first to jointly capture anticipated regret and level-*k* thinking to analyze bidding outcomes in multistage auctions both theoretically and empirically. The modeling framework and analysis allow us to derive structural insights that are generalizable to other markets and products beyond the specific ones considered in this paper. They also provide the analytical foundation to enable a successful field implementation that is shown to generate positive impacts in practice.

A third related stream of literature studies auctions in the field (see Ockenfels et al. 2021 for a recent review). This literature focuses predominantly on markets in developed economies and examines how various information disclosure affects auction outcomes. For example, Lucking-Reiley et al. (2007) conduct a field experiment on eBay by auctioning collectible trading cards and find that, in contrast to the revenue equivalence theorem, descending-price auctions result in higher revenues than first-price sealed-bid auctions. Tadelis and Zettelmeyer (2015) show that in wholesale auto auctions, disclosing quality information of the cars helps to better match heterogeneous buyers with cars of varying quality, thus improving market efficiency. Lu et al. (2019) conduct a field experiment in the Dutch flower auction market and find that concealing winners' identities can significantly increase the average winning price in sequential auctions. Our contributions to this literature are twofold. First, we expand the literature to consider field-based auction design in developing markets with constrained resources and distinct operational characteristics. Second, whereas most works in the above literature focus on testing specific theoretical predictions with field data, our main goal is to design an *implementable* auction mechanism to increase revenue for farmers. Thus, we introduce a multimethod approach for practical and effective auction design guided by both operational and behavioral insights. Such an approach can be generalized to other real-world market design settings to enhance the impact of auction research in practice.

The remainder of the paper is organized as follows. Section 3 presents the operational backgrounds and characteristics of the UMP that lead to the proposal of a two-stage auction. Section 4 examines the market environment and the traders' bidding behavior in the two-stage auction based on semistructured interviews in the field. Section 5 develops and analyzes a behavioral auction model motivated by the field insights. Section 6 describes the implementation of the two-stage auction in a major market of tur, empirically evaluates its impact on farmers' revenue, and examines the relevance of the behavioral insights with postimplementation field data. Section 7 concludes the paper.

3. Operational Considerations and the Design of the Two-Stage Auction

To fully appreciate the operational context, we first describe the current trading process on the UMP. Under the Agricultural Produce Market Committee (APMC) Act that regulates trading of certain major commodities (e.g., grains and lentils) in the state, all agricultural trades of these commodities between farmers and traders must be conducted in one of the regulated markets through the UMP. At the start of a day, farmers bring their products to a commission agent of their choice in a local market. All lots arriving to a market are posted on the UMP and are visible to all traders. Throughout the day, traders or their representatives visit commission agent shops to identify lots they want to purchase and the prices to bid. Afterward, they must submit their (private) bids for all the lots they want to purchase on the UMP before a preannounced cutoff time (currently 2:30 p.m.). Once the bidding window is closed, the highest bidder for each lot is declared as the winner. The winner pays the farmer the highest bid, and the winning bid is announced publicly. That is, the current auction design on the UMP is a single-stage, first-price, sealed-bid auction at an individual lot level.

To identify potential alternatives to the current design, we begin by evaluating the feasibility of alternative auction mechanisms commonly used in other settings. One particularly relevant analysis is that of eNAM launched by the central government of India, due to the close similarities in market operations and cultural characteristics between the eNAM and the UMP. eNAM uses ascending auctions to determine market prices. In an ascending auction, traders sequentially increase their bids on a lot until only one trader remains. This last trader wins the lot and pays the current highest bid. However, traders initially participating in eNAM complained that having to constantly monitor and potentially adjust bids for hundreds of lots involved too much effort, seriously disrupted other aspects of their operations (e.g., arranging postauction logistics), and became operationally infeasible during peak seasons. As a result, trader participation on the eNAM was not sustained, and the platform has not realized its intended impact to date. In light of such participation constraints on the traders' side, the Karnataka government was not supportive of adopting an ascending auction.

A similar analysis concerns the farmers' perspective. Transitioning from the traditional markets to the UMP involved substantial efforts to build trust with the farmers. Therefore, the government was very mindful of maintaining this hard-earned trust. As a result, the government rejected the design of a second-price sealed-bid auction because it would be an immense challenge to explain to farmers why they are paid the *second* highest bid, but not the top bid. These and other in-depth discussions with the UMP officials have ruled out the operational feasibility of several theoretically appealing and commonly used auction designs (Table 1).

A number of critical operational requirements result from this first phase of the design process. In particular, the new auction design must minimize disruption to the current trading and postauction processes; be easy to explain to farmers and traders, as well as satisfy their participation constraints; and require minor technical changes on the platform to accommodate resource constraints. Taking these factors into account, one promising mechanism is the multistage auction with qualification requirements. This mechanism is commonly used in selling high-value assets—for example, in mergers and acquisitions. In a multistage auction, sealed bids are solicited in multiple stages. In each stage, a fraction of the bidders are eliminated, according to certain predefined qualification requirements, and some information about the current bids is disclosed. As the stages progress, bids increase, and, eventually, the asset is sold to the bidder with the highest bid. This design is akin to

Table 1. Commonly Used Price Discovery Mechanisms and Their Operational Feasibility on the UMP

Mechanism	Used by	Operational feasibility on the UMP
Ascending or descending auction	eNAM, India (Reddy 2018); Dutch flower markets (Roth and Ockenfels 2002)	Requires substantial trader efforts to monitor bids for many lots; risk of last-minute bidding crashing the platform; requires significant updates to the platform
Second-price auction	Online advertising exchange (Golrezaei et al. 2017)	Hard to get farmers' buy-in for receiving the second highest bid (as opposed to the highest bid); requires significant training for the traders
Combinatorial auction	Chilean school meal auction (Kim et al. 2014)	Too complex to implement or explain to market participants; requires significant training and updates to the platform
Double auction	Kudu, Uganda (Newman et al. 2018)	Farmers lack information, financial, and market power to give competitive ask prices; insufficient computers for all farmers; requires significant updates to the platform
Posted price	ITC e-Choupal, India (Anupindi and Sivakumar 2007)	UMP is not a buyer, but a platform to enable trades
Warehouse-based sales	Ethiopia Commodity Exchange (Hernandez et al. 2017)	Lack of infrastructure regarding warehouse storage and quality grading
Multistage auction with qualifications	Real estate market, mergers and acquisitions (Ye 2007)	Similar to conducting multiple first-price sealed-bid auctions, easy to explain to market participants, and requires incremental updates to the platform

conducting multiple first-price sealed-bid auctions and, thus, has a strong implementation appeal to the government. In the end, the simplest form of a multistage auction—that is, a two-stage auction with a qualification stage—is proposed.

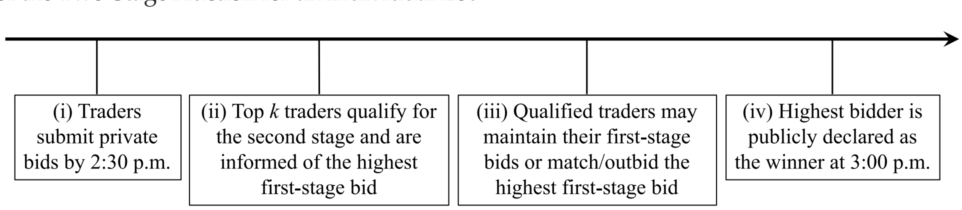
Figure 2 presents the timeline and rules of the proposed two-stage auction mechanism. Specifically, the auction is conducted at an individual lot level as before. In the first stage, all traders bid as in the current first-price sealed-bid auction. At the end of the first stage, the current highest bid of each lot is disclosed to the top k traders (based on their first-stage bids) who have bid on the lot. These traders are "qualified" to bid again in the second stage, and they are not allowed to withdraw their first-stage bids. They can either maintain their first-stage bids, or, if they want to change, they must at least match and potentially outbid the highest first-stage bid. The second-stage bidding is to be completed within a much shorter time window than the first stage. The top bidder at the end of the second stage is declared as the winner.[2] In this design, the first stage serves as a qualification stage, and the second stage is essentially another first-price sealed-bid auction with a reserve price among the subset of qualified traders.

Remark. It is important to clarify that the operational considerations discussed here reflect the system constraints at the time of the research. It should not be viewed as ruling out the feasibility of the other mechanisms indefinitely. Nevertheless, our interactions with government officials indicate that systemic infrastructural changes will likely take a long period of time to realize. Therefore, identifying alternatives that can benefit farmers in the more immediate future is highly valuable. As the UMP and its associated physical infrastructure continue to develop, we expect that some of the currently infeasible mechanisms may become viable. For example, the government is actively investing in quality grading and storage facilities, which can make a warehouse-based sales mechanism a potentially appealing option in the future.

4. Traders' Bidding Behavior

Changing the price discovery mechanism on the UMP can have a profound impact on farmers' revenue. Because two-stage auctions have never been used in agricultural markets, we must carefully evaluate their viability to enhance farmers' revenue. Therefore, the next step in the design process is to examine the traders' bidding behavior in the proposed two-stage auction. To do so, we utilize semistructured field interviews to address the following two aspects: (i) determine the

Figure 2. Rules of the Two-Stage Auction for an Individual Lot



right auction model to suitably capture the field setting, and (ii) examine the relevance of various behavioral regularities in affecting the traders' bidding strategies.

Our semistructured field interviews were conducted with a group of tur traders in a major lentils market. The design of the interview follows well-established principles in the social sciences (Stantcheva 2022; see Online Appendix O.7). To conduct the interviews, one of the authors visited the market for four days on a randomly selected week during the season and interviewed all available traders. Traders who did not come to the market during those days were not interviewed. Importantly, traders were not informed of the interview in advance. Hence, potential selection problems on the interviewee side are likely to be minimal. In total, we interviewed 13 traders (out of 19 major traders in the market). These traders have, on average, 16 years of experience in agricultural trading and have won over 68% of all the lots (70% of the total quantity) traded in the pretreatment data. Therefore, the sample represents a majority of the traders whose bids substantially influence the final market prices. Each interview took about an hour to complete.[3]

4.1. The Auction Setting

The first set of questions in the interview help us determine a suitable auction model to properly capture key elements in the field setting.

4.1.1. Common-Value vs. Private-Value Auction. We first examine whether private-value or common-value auctions would be more representative of the field setting. If there were sufficient uncertainty in the quality of a lot, then a model with an unknown common-value component would be more appropriate. Contrary to this hypothesis, however, the traders interviewed explain that there is little uncertainty about the lots' quality once they have physically inspected the lots before the first-stage bidding. This is because the primary factors determining the quality of tur are observable lot-level features, such as seed size, moisture level, and foreign particles in the lot, and the traders are all highly experienced in quality inspection. In addition, the traders also talk to the farmers directly to further confirm their assessment. Because the farmers and traders have been trading in these markets for generations, it is in the farmers' best interest to truthfully reveal their lots' quality. Therefore, after the inspection and their conversations with the farmers, all traders would know the quality of a lot with high certainty, and they also know that other traders bidding for the lot would have similar quality assessment. These insights suggest that private-value auction models are more representative of our setting.

4.1.2. What Drives the Traders' Private Valuations of a Lot? To answer this question, it is instructive to understand the process of tur trading from a trader's perspective. Tur is an important ingredient in the Indian diet, and it is sold for retail consumption across the country. The raw tur sold on the UMP first needs to be processed in mills before reaching the retail market. Tur mills are spread across India and owned by a diverse set of mill owners. The traders on the UMP act as intermediaries to source raw tur for these mill owners. The transactions between the traders and their buyers (i.e., mill owners) work as follows. In the morning of a trading day, a trader talks to their buyer to agree on the per kilogram (quality-dependent) price for that day. Therefore, from the trader's perspective, there is no uncertainty in the resale price once quality is determined. Nonetheless, these resale prices can vary across different traders, primarily due to idiosyncratic differences across the mill owners to whom they sell. Each trader also has their own target margin. The agreed-upon resale price minus the margin would determine the trader's private valuation of a lot (given quality).[4]

4.1.3. Single-Unit vs. Multiunit Auction. Because retail demand for tur is consistently high, mill owners are, in general, supply constrained and are willing to buy as much raw tur as possible from the traders. Furthermore, the traders commented in the interviews that they do not have budget constraints, given the supply constraints. Hence, although each trader bids for multiple lots every day, the seemingly multiunit auction can be reasonably decoupled into multiple independent single-unit auctions.

Taken together, the above field insights establish that the setup of a single-unit independent private-value auction is a reasonable model for our context.

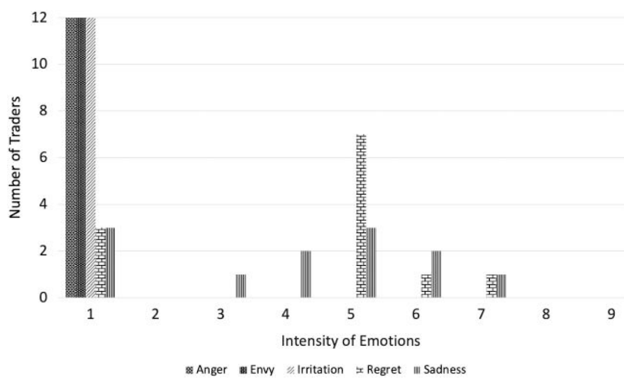
4.2. Traders' Behavioral Regularities

As shown in the experimental auction literature (Kagel and Levin 2016, Elmaghraby and Katok 2018), human bidders are often affected by behavioral regularities and deviate from equilibrium bidding outcomes. Hence, it is important to understand what behavioral factors might be salient and affect the traders' bidding strategies in the two-stage auction. Although many behavioral factors may plausibly play a role, we must carefully select which factors to focus on, given realistic time constraints of the interviews (i.e., most traders only had about an hour to participate).[5] Thus, we choose to focus on behavioral factors that satisfy two conditions: (i) it has been well documented to explain bidding behaviors in the literature; and (ii) it can become salient given the two-stage auction setup. In addition, we carefully discuss our choices with our field collaborators to ensure that the interview questions would make sense to the traders, and, hence, we can obtain useful responses in the interviews. Based on these considerations, we specifically investigate two factors: anticipated regret and competitive arousal.

4.2.1. Anticipated Regret. The effect of anticipated regret on bidding behavior has been observed extensively in laboratory experiments of single-stage auctions (Filiz-Ozbay and Ozbay 2007, Engelbrecht-Wiggans and Katok 2008). Bidders are shown to bid higher prices if they know that the winning bid will be disclosed to them when they lose (versus no disclosure). This is because bidders anticipate feeling regret ex post that they could have won and made a profit. To avoid such regretful emotions, bidders may bid more aggressively ex ante.[6] We expect a priori that, as compared with the single-stage first-price auction, the two-stage auction would *intensify* the traders' anticipated regret for two possible reasons. First, traders are exposed to "repetitive" regret because information feedback of both not qualifying for the second stage and losing the auction is made salient. According to evidence in clinical psychology (Coricelli et al. 2005, Roese et al. 2009), regretful emotions would intensify as a result. Second, for a given bid, traders have a higher chance to qualify for the second stage in the two-stage auction than to win in a single-stage first-price auction. Research has shown that anticipated regret is more intense for (fearing of) missing out on desired outcomes that are more likely to happen (Robinson and Botzen 2019). To summarize, we expect that the traders' stronger anticipated regret in the two-stage auction would motivate them to increase their first-stage bids over their bids in the single-stage first-price auction, thereby benefiting the farmers.

To examine the presence of anticipated regret among the traders in a two-stage auction, we design the following test in the interviews. The traders were presented a hypothetical scenario of a two-stage auction, which states the following: "Assume that there are 15 traders in the market today and you participate in the [two-stage] auction. In particular, only top 3 bidders in the first round will qualify for the second-round bidding."[7] Given this hypothetical information, the traders were asked to state their first-stage bids. Afterward, they were told "Suppose at the end of the first round, you find that you did not qualify for the second round. Please rate the intensity of the emotions listed below [anger, envy, irritation, regret, sadness] you would experience in that situation." A higher score (ranging from 1 to 9) indicates a stronger feeling of the corresponding emotion, with 1 meaning no feeling at all and 9 meaning an extremely strong feeling. The design of these questions follows from Filiz-Ozbay and Ozbay (2007). Figure 3 presents the distribution of the stated intensity of different emotions across traders.[8] We observe that a considerable number of traders indeed feel substantial regret for not being qualified for the second stage. Thus, the results provide evidence in support of our conjecture; that is, (at least some of) the traders would likely bid more aggressively in the first stage of the two-stage auction due to strong anticipated regret.[9]

Figure 3. Trader Interview: Distribution of Stated Intensity of Different Emotions Under Nonqualification



Note. A higher score indicates a stronger feeling of the corresponding emotion: 1 = no feeling at all, and 9 = an extremely strong feeling.

4.2.2. Competitive Arousal. In the two-stage auction, the number of bidders would be substantially reduced in the second stage to only allow a small group of qualified traders to participate. This setup can make another behavioral factor—namely, competitive arousal—particularly salient. Researchers have shown that when perceptions of rivalry and time pressure are heightened, human decision makers may be aroused to form a "win-at-any-cost" mindset, which can often result in costly mistakes (Malhotra 2010, Adam et al. 2015). In the second stage of the two-stage auction, there are only a few rivals, and traders need to bid in a short time frame. These features can elevate the influence of competitive arousal among the qualified traders. As a result, qualified traders may bid aggressively in the second stage to ensure that they win the qualified lots, thus improving the farmers' revenue.

To test the relevance of competitive arousal, the traders were provided a similar hypothetical two-stage auction scenario described earlier, with two key differences. First, the traders were asked to imagine that they qualified for the second round and were provided a hypothetical top bid from the first round. Second, the number of qualified bidders in the second round was either three or seven, and the time allowed for the second-stage bidding was either four or 15 minutes. The traders were asked to indicate their second-stage bids for all four scenarios. The theory of competitive arousal would predict that the traders would bid more aggressively when a smaller number of traders qualified and/or when they were given a shorter time to bid (i.e., when the perception of rivalry was heightened).

Contrary to this prediction, the traders' second-stage bids were at most 0.01% higher than the disclosed highest first-stage bid across all scenarios. That is, contrary to competitive arousal, which predicts that the traders would become *more* competitive in the second round due to a heightened sense of rivalry, we find that the traders only added minimal increment to the first-stage top bid.

5. A Behavioral Auction Model

In this section, we build on the field and behavioral insights in Section 4 to develop and analyze a parsimonious, yet behaviorally plausible, auction model. Our main goal is to utilize the model to examine *when* the proposed two-stage auction may outperform the legacy single-stage first-price sealed-bid auction to generate higher revenues for the farmers. We remark that the model is stylized in nature and meant to help us gain structural insights into the traders' bidding strategies and the resulting auction revenue. Thus, we focus on capturing the first-order factors, while simplifying away secondary aspects of the actual complex bidding environment.

In particular, based on our discussion in Section 4.1, we consider an independent private-value auction, in which three traders bid for a single lot. Each trader's valuation of the lot is modeled as $q + v$, where q is the objective quality of the lot commonly known to all three traders, and v is each trader's private value. Without loss of generality, this private value is independently drawn from a uniform distribution with support $[0, 1]$. In the case of the two-stage auction, the top two traders in the first stage will qualify to bid again in the second stage, according to the rules specified in Figure 2. Given our findings in Section 4.2.1, we consider traders to be influenced by anticipated regret when participating in the auction. Furthermore, the strength of anticipated regret is intensified in the two-stage auction compared to the single-stage first-price auction.

We next introduce the solution concept for our analysis. Specifically, we analyze the traders' bidding strategies under the nonequilibrium framework of "level-k" thinking, instead of solving for a Bayesian-Nash equilibrium. This choice is motivated by extensive empirical evidence that the level-k thinking framework can better predict human behaviors than canonical equilibrium models do (Crawford 2003, Camerer et al. 2004, Costa-Gomes and Crawford 2006, Crawford and Iriberri 2007, Crawford et al. 2013). In particular, Crawford and Iriberri (2007) find strong behavioral support of level-k thinking with data from multiple auction experiments. The fundamental idea of level-k thinking is that people do not perform infinite iterations of best-response reasoning; instead, they engage in a limited number of such iterated thinking. An appealing property of level-k thinking is that it nests the equilibrium framework as a special case. Specifically, when people can perform a sufficiently high level of strategic thinking, the predictions of the level-k thinking framework "converge" to equilibrium predictions. Therefore, level-k thinking can be viewed as a more general solution concept that imposes less strict assumptions on the depth of strategic reasoning performed by humans.

Formally, the level-k model assumes that bidders' behaviors follow a hierarchy of bidding rules, given their "types." The type of a bidder captures the bidder's level of sophistication in performing iterated best-response reasoning against other bidders' strategies. The hierarchical bidding rules stipulate the following structure: (i) type L_0 bidders are nonstrategic; (ii) type L_k bidders assume all other bidders follow the bidding rule of type L_{k-1} and best respond to that rule. A higher level in the hierarchy (i.e., a larger k) corresponds to more sophisticated reasoning. Note that if type L_k bidders bid according to an equilibrium strategy, then all type L_m bidders with $m > k$ will also employ the equilibrium strategy; that is, they have "converged" to an equilibrium. This is because, by definition, an equilibrium strategy is a best response to itself. In the literature, L_0 bidders are typically assumed to bid either their true valuations or uniformly randomly over the support of the bidders' value distribution (Crawford and Iriberri 2007). In our context, we consider L_0 bidders to bid their true valuations because it is unlikely that highly experienced professionals such as the traders would bid randomly. Furthermore, we consider the most sophisticated traders to be of type L_2. This is reasonable because researchers have shown across an extensive set of strategic decision contexts, including auctions, that types of L_3 or higher are rarely observed (Camerer et al. 2004, Crawford and Iriberri 2007, Crawford et al. 2013). In what follows, we analyze the bidding strategies of L_1 and L_2 traders in both the first-price sealed-bid auction and the two-stage auction. To simplify exposition, we normalize the objective lot quality, q, to be zero.

We first analyze the traders' bidding strategies in the first-price sealed-bid auction when the traders are influenced by anticipated regret. The bidding strategy of a type L_k trader can be characterized by a mapping $B_k^{\mathrm{FP}}(v) : [0, 1] \to \mathcal{R}^+$, which specifies the trader's bid given his valuation v. Type L_0 traders bid their valuation v truthfully by our assumption; that is, $B_0^{\mathrm{FP}}(v) = v$. We next consider a type L_1 trader's bidding strategy. A type L_1 trader assumes that all other traders are of type L_0 and bid their true valuations. Let V_1 and V_2 be the valuations of the other two traders, and let v and b be the focal trader's valuation and bid. If the trader wins (i.e., when $b > \max\{V_1, V_2\}$), his payoff would be $(v - b)$. Conversely, if the trader loses (i.e., when $b \le \max\{V_1, V_2\}$), he earns a zero payoff and would also feel regret if he could have won with a positive payoff—that is, if $(v - \max\{V_1, V_2\}) > 0$. Following the literature (Filiz-Ozbay and Ozbay 2007, Engelbrecht-Wiggans and Katok 2008), we model the magnitude of this regret to be proportional to the forgone revenue—that is, $(v-\max\{V_1, V_2\})^+$, where $x^+ = x$ if $x > 0$ and zero otherwise. Summarizing the above, a type L_1 trader's expected payoff given his valuation v and bid b can be characterized as follows.[10]

$$\pi_1^{\mathrm{FP}}(v,b) = (v-b)E[\mathbb{1}\{b > \max\{V_1, V_2\}\}] - E[\lambda_L (v - \max\{V_1, V_2\})^+ \mathbb{1}\{b \le \max\{V_1, V_2\}\}], \tag{1}$$

where $\lambda_L \ge 0$ is the regret parameter that captures the importance of regret relative to the monetary payoff in the trader's expected payoff, $\mathbb{1}\{A\}$ denotes the occurrence of the probabilistic event A, and the expectation is taken on the other two traders' true valuations V_1 and V_2. The trader chooses a bid b to maximize his expected payoff in Equation (1). The following result summarizes a type L_1 trader's bidding strategy in the first-price sealed-bid auction.

Proposition 1. *In the first-price sealed-bid auction, the bidding strategy of a type L_1 trader with valuation v is $B_1^{\mathrm{FP}}(v) = \frac{2+2\lambda_L}{3+2\lambda_L} v$ for all $v \in [0,1]$.*

Filiz-Ozbay and Ozbay (2007) show that if bidders are subject to anticipated loser's regret, then the bidding strategy in a symmetric pure strategy Bayesian-Nash equilibrium in the first-price sealed-bid auction takes precisely the same form as the one in Proposition 1. That is, a type L_1 trader's bidding strategy has converged to the equilibrium strategy. Because a type L_2 trader assumes all other traders are of type L_1 and best responds to the bidding strategy in Proposition 1, we know that the type L_2 trader must employ the exact same bidding strategy, as summarized in the following corollary.

Corollary 1. *In the first-price sealed-bid auction, the bidding strategy of a type L_2 trader is exactly the same as that of a type L_1 trader; that is, $B_2^{\mathrm{FP}}(v) = \frac{2+2\lambda_L}{3+2\lambda_L} v$ for all $v \in [0,1]$.*

We next analyze the traders' bidding strategies in the two-stage auction. Let $B_k^{Sj}(v)$ be the bidding function of a type L_k trader in stage j. We solve for the traders' bidding strategies by backward induction, starting from their second-stage bidding. Let B_{top} be the highest first-stage bid. First, consider a type L_1 trader. There are two scenarios. First, if the type L_1 trader is ranked first, then he believes that the other two traders' valuations are both below B_{top} (recall that a type L_1 trader assumes all other traders are of type L_0 and bid truthfully). Hence, the trader will not change his bid. Second, if the type L_1 trader is instead ranked second, then he believes that the highest valuation among the other two traders is equal to B_{top}. Hence, if the trader's own valuation is greater than B_{top}, then he would increase his bid to $B_{\mathrm{top}} + \epsilon$ to win the auction, where ϵ corresponds to the smallest practical increment. Otherwise, he would not change his bid. These results are summarized in the following lemma.

Lemma 1. *In the second stage of the two-stage auction,*

(i) *If a type L_1 trader is ranked first, then he would not change his bid—that is, $B_1^{S2}(v) = B_{\mathrm{top}}$.*

(ii) *If a type L_1 trader is ranked second, then he bids $B_1^{S2}(v) = B_{\mathrm{top}} + \epsilon$ for a small $\epsilon > 0$ if $v > B_{\mathrm{top}}$, and he does not change his bid if $v \le B_{\mathrm{top}}$.*

Now, consider a type L_1 trader's bidding strategy in the first stage. This trader again assumes the other two traders are of type L_0 and bid their true valuations V_1 and V_2 in the first stage. There are three scenarios. First, if the type L_1 trader bids $b > \max\{V_1, V_2\}$, then he would rank first in the second stage. Based on Lemma 1(i), he would not change his bid in the second stage and expects to earn a payoff of $(v-b)$. Second, if the type L_1 trader bids $b \in (\min\{V_1, V_2\}, \max\{V_1, V_2\}]$, then he would rank second in the second stage. Based on Lemma 1(ii), if his valuation $v > \max\{V_1, V_2\}$, then he would slightly outbid $\max\{V_1, V_2\}$ and expects to earn a payoff of $(v - \max\{V_1, V_2\})$;[11] if, instead, $v \le \max\{V_1, V_2\}$, then he would not change his bid and expects to lose the auction with a zero payoff. Finally, if the type L_1 trader bids $b < \min\{V_1, V_2\}$, then he would rank third and would not qualify for the second stage. As shown in Section 4.2.1, the trader would experience (nonqualification) regret in this case if he could have qualified and then won the auction with a positive expected payoff. Similar as before, we model the magnitude of this regret to be proportional to the forgone revenue—that is, $(v - \max\{V_1, V_2\})^+$—and we use the parameter $\psi \ge 0$ to denote the importance of this regret relative to his monetary payoff. Summarizing the above three scenarios, the expected payoff of a type L_1 trader with valuation v and bid b in the first stage of the two-stage auction can be characterized as follows.

$$\begin{aligned}\pi_1^{S1}(v,b) = {} & (v-b)E[\mathbb{1}\{b > \max\{V_1, V_2\}\}] \\ & + E[(v - \max\{V_1, V_2\})^+ \mathbb{1}\{\min\{V_1, V_2\} < b \\ & \le \max\{V_1, V_2\}\}] \\ & - \psi E[(v - \max\{V_1, V_2\})^+ \mathbb{1}\{b \le \min\{V_1, V_2\}\}].\end{aligned} \tag{2}$$

The trader chooses b to maximize this expected payoff. The following proposition characterizes the type L_1 trader's bidding strategy in the first stage of the two-stage auction.

Proposition 2. *In the first stage of the two-stage auction, the bidding strategy of a type L_1 trader with valuation v is $B_1^{S_1}(v) = \frac{\psi - \sqrt{3+2\psi}}{\psi - 3} v$ for all $v \in [0,1]$.*

Finally, we analyze a type L_2 trader's bidding strategy in the two-stage auction. The following lemma first characterizes a type L_2 trader's bidding strategy in the second stage.

Lemma 2. *In the second stage of the two-stage auction,*

(i) *If a type L_2 trader is ranked first, then he bids $B_2^{S2}(v) = B_{\text{top}} + 2\epsilon$ for a small $\epsilon > 0$.*

(ii) *If a type L_2 trader is ranked second, then he bids $B_2^{S2}(v) = B_{\text{top}} + \epsilon$ for a small $\epsilon > 0$ if $v > B_{\text{top}}$, and he does not change his bid if $v \le B_{\text{top}}$.*

The reasoning behind this result follows similarly to our analysis of a type L_1 trader's bidding strategy in the second stage. Specifically, a type L_2 trader assumes all other traders are of type L_1 and bid according to Lemma 1 and Proposition 2 in the second and first stage, respectively. There are two scenarios. First, if the type L_2 trader is ranked second, then, based on Lemma 1(i), he believes that the first-ranked trader would bid B_{top} again. Hence, the type L_2 trader can win the auction by slightly outbidding the type L_1 trader, and he will do so if the resulting payoff is positive. This scenario is summarized in Lemma 2(ii). Second, if the type L_2 trader is instead ranked first, then, based on Lemma 1(ii), he believes that the second-ranked trader would bid $B_{\text{top}} + \epsilon$ if their valuation is greater than B_{top}. Otherwise, the second-ranked trader would not change his bid. Lemma 2(i) shows that in this case, the type L_2 trader's best response is to bid $B_{\text{top}} + 2\epsilon$ for a small ϵ.

Now, consider a type L_2 trader's bidding strategy in the first stage. The trader assumes that the other two traders are of type L_1 and bid their bidding strategy $B_1^{S_1}(.)$ from Proposition 2 in the first stage. Following the same reasoning as for the type L_1 trader in the first stage, the type L_2 trader's expected payoff, given his valuation v and bid b, can be characterized as follows.

$$\begin{aligned}\pi_2^{S1}(v,b) &= (v-b)E[\mathbb{1}\{b > \max\{B_1^{S_1}(V_1), B_1^{S_1}(V_2)\}\}] \\ &\quad + E[(v - \max\{B_1^{S_1}(V_1), B_1^{S_1}(V_2)\})^+ \\ &\quad \mathbb{1}\{\min\{B_1^{S_1}(V_1), B_1^{S_1}(V_2)\} < b \\ &\quad \le \max\{B_1^{S_1}(V_1, B_1^{S_1}(V_2)\}\}] \\ &\quad - \psi E[(v - \max\{B_1^{S_1}(V_1), B_1^{S_1}(V_2)\})^+ \\ &\quad \mathbb{1}\{b \le \min\{B_1^{S_1}(V_1), B_1^{S_1}(V_2)\}\}],\end{aligned}$$

where $B_1^{S_1}(\cdot)$ is defined in Proposition 2. The next result characterizes the type L_2 trader's bidding strategy in the first stage.

Proposition 3. *In the first stage of the two-stage auction, the bidding strategy of a type L_2 trader with valuation v is $B_2^{S1}(v) = \frac{\psi - \sqrt{3+2\psi}}{\psi - 3} v$ for all $v \in [0,1]$.*

Comparing the results for the first-price sealed-bid auction (Proposition 1 and Corollary 1) with those for the two-stage auction (Propositions 2 and 3), we can show the following result.

Proposition 4. *There exists a threshold $\overline{\lambda}_H > \lambda_L$ such that if $\psi \ge \overline{\lambda}_H$, then the expected revenue under a two-stage auction will be higher than that under a single-stage first-price sealed-bid auction.*

Lemmas 1 and 2 suggest that once a trader qualifies for the second stage, they would only slightly outbid the highest first-stage bid to win the auction. These results are consistent with our findings in the trader interviews (see Section 4.2.2), where we found that the traders only added minimal increment to the first-stage top bid in the second stage. Conversely, the presence of anticipated regret motivates a trader to bid more competitively in the first stage. Note that qualifying for the second stage requires a lower bid than winning the auction in the single-stage first-price sealed-bid auction. Thus, only when the intensified anticipated regret in the two-stage auction is sufficiently strong can the more competitive bidding in the first stage counteract the limited bid increase in the second stage to yield a higher expected revenue in the two-stage auction than in the single-stage first-price auction.

An immediate implication from this proposition is that implementing the two-stage auction would be beneficial for those commodities for which the traders tend to exhibit strong anticipated regret. This is likely to be the case for commodities whose supply is constrained relative to market demand or those that are of high economic values. This result strengthens our confidence that the two-stage auction would benefit tur farmers, because tur, as well as other lentils, have limited supply relative to the consistently high consumption across the country due to their importance in the mostly vegetarian Indian diets. In addition, the analysis suggests that high-value cash crops, such as cotton and turmeric, are also good candidates for implementing the two-stage auction.

6. Field Implementation and Impact Assessment of the Two-Stage Auction

6.1. Field Implementation

The market in which the interviews were conducted was chosen to be the treatment market for implementing the two-stage auction.[12] This choice is driven by four key reasons. First, there exists another similar market that can serve as the control market to allow for empirical evaluation of the impact of the implementation on market prices. Second, the field interview data and the lot-level auction data from the implementation can later be combined to provide further observational insights regarding the traders' bidding behavior. Third, this market is a major lentils market with sufficient trading quantities to allow for impact assessment. Fourth, the government wanted to experiment with the new auction design in a market where they believed resistance to change from market participants was relatively low, based on their earlier experience of launching the UMP in the first place. The last consideration was to ensure that even if the implementation did not go as planned, the government's goodwill could still be salvaged.

To ensure a successful launch, extensive training and discussions were carried out with the market participants weeks before the implementation. Particularly to get the traders' buy-in, the two-stage auction design was "marketed" to them as giving them an additional chance to win the lots that they would have otherwise lost. A number of implementation details resulted from interactions with the traders during the training phase. First, the traders requested that the length of the second stage should be no more than half an hour because they had to perform a series of postauction tasks by the end of the day (e.g., weighing, cleaning, packing, and transporting the lots, transferring payments to the farmers). Second, considering the potential average number of lots that a trader would be qualified to bid again (based on historical data) and the feasibility of doing so within half an hour, only the top three bidders for each lot would be qualified—that is, $k = 3$ in Figure 2. Third, the traders requested that for lots for which they qualify to bid again, their own rank among the top three should also be disclosed (in addition to the highest first-stage bid) to allow for more informed bidding in the second stage.

After accounting for this feedback, the final implementation was determined as follows: (i) the cutoff time for the first-stage bidding would remain at 2:30 p.m., as in the current process; (ii) at that time, the traders would receive mobile text messages that inform them for which lots they qualify for the second stage; (iii) the second-stage bidding would begin at 2:35 p.m.; (iv) the traders would observe on the UMP the highest first-stage bids and their own rank for the lots for which they qualify, and they have until 3:00 p.m. to increase their bids; and (v) at 3:00 p.m., the winners for *all* lots in the market would be declared. A single winner declaration time for all lots is chosen to simplify the coordination of postauction activities.[13] The implementation was launched on February 22, 2019.

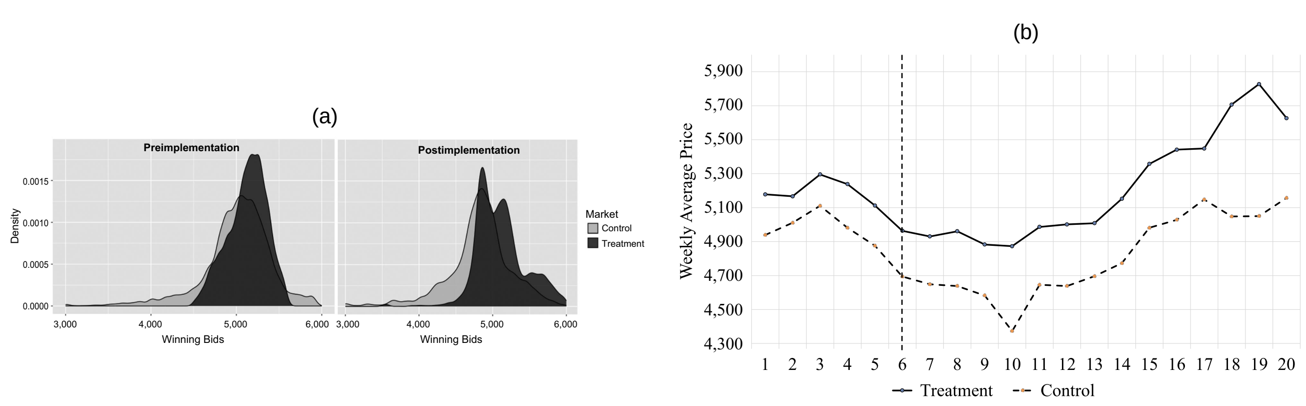
To empirically evaluate the impact of implementing the two-stage auction on farmers' revenue, we adopt a difference-in-differences (DID) approach (Imbens and Wooldridge 2009). The control market is selected based on the following criteria: (i) it must be a major market of tur, as is the treatment market;[14] (ii) there are a similar number of traders in the two markets; (iii) the prices in the two markets preimplementation must satisfy the parallel trends assumption; (iv) the two markets are sufficiently far apart (223 kilometers away) that no traders or farmers would switch to the other market for trading, as confirmed by market officials; and (v) domain knowledge from the collaborators further confirms that the two markets are similar in terms of infrastructure, the quality of the lots, and the characteristics of the traders.

The empirical analysis utilizes lot-level auction data for the control and treatment markets from one month prior to the implementation until May 31, 2019.[15] The data contain all of the bids that were placed on the UMP within this time window in both markets (a total of 172,342 bids), including all bids placed in both stages after the two-stage auction was implemented in the treatment market. Note that the bids placed are unit prices in rupees per quintal (i.e., 100 kilograms). In all of our subsequent discussion, all bids and prices referred to are unit prices. Online Appendix O.2 presents the summary statistics of our data. Figure 4(a) illustrates the distribution of the winning bids in the control (light gray) and treatment (dark gray) markets preimplementation (left) and postimplementation (right). Figure 4(b) shows the time series of the average weekly prices in these two markets during the time window of our analysis. The vertical line indicates the date on which the two-stage auction was implemented in the treatment market. We observe from both figures that the price difference between the two markets increases after the implementation. We next present the details of the formal impact assessment.

6.2. Comparability of the Treatment and Control Markets

We first verify the comparability of the two markets along a few important market features. Table 2 presents the

Figure 4. Comparing Prices in Treatment and Control Markets Preimplementation and Postimplementation



Notes. (a) Distribution of winning bids. Panel (a) presents the distribution of winning bids at the lot level in the control and treatment markets preimplementation (left) and postimplementation (right). (b) Average weekly prices. Panel (b) presents the time series of the average weekly prices in these two markets. The vertical line indicates the date on which the two-stage auction was implemented in the treatment market.

Table 2. Summary of Demographic and Market Features in the Treatment and Control Markets Preimplementation

Variable	Treatment	Control
Total Population (in millions)	1.7	1.9
Total Households (in thousands)	320	364
Share of Population (Rural)	75%	75%
Share of Population (SC/ST)	37%	40%
Literacy Rate	79%	70%
Poverty Rate	28%	22%
Households with Electricity	96%	95%
Households with Improved Sources of Drinking Water	96%	89%
Number of Traders[a]	25.50	24.67
	(3.5)	(7.6)
Number of Bids per Lot	5.10	5.52
	(0.46)	(0.83)
Unit price (rupees/100 kg)	5,165	4,943
	(106)	(136)

Notes. Demographic features are based on 2011 census data and obtained from https://niti.gov.in. SC and ST in row (4) refer to Scheduled Castes and Scheduled Tribes, which are historically discriminated castes in India.

[a]Values presented for the last three market features are means (standard deviations). Results from t-tests show that there is no statistically significant difference in the average number of traders and the average number of bids per lot between the two markets. Conversely, the average market prices between the two markets are significantly different ($p = 0.014$).

summary statistics related to tur trading in the control and treatment markets, as well as demographic features for the regions where these markets are located based on administrative data from NITI Ayog. We observe from the table that the two markets are comparable in these features prior to the implementation. Furthermore, domain knowledge from the collaborators also assures that the two markets are both major markets of tur with similar characteristics in terms of market infrastructure, the quality of the lots, and the characteristics of the traders.

6.3. Parallel Trends Assumption

The key identifying assumption for a DID approach is that the difference in market prices between the treatment and control markets would have remained constant over time in the absence of the treatment; that is, the price trends are parallel between the two markets prior to the treatment. Following established methods in the literature (Autor 2003, Angrist and Pischke 2008), we estimate the following model using pretreatment data to verify the parallel trends assumption:

$$\log(P_{m,t,l}) = I_{m,l} \times \left(\sum_{w=1}^{5} \gamma_w M_{w,l}\right) + \delta_1 \log(Q_{m,t,l}) + \delta_2 \log(N_{m,t}) + \delta_3 \log(S_{m,t}) + \delta_4 R_{m,t} + \delta_5 B_{m,t,l} + (\text{market, date fixed effects}) + \epsilon_{m,t,l}. \tag{3}$$

The variable $P_{m,t,l}$ is the winning bid (in rupees/100 kilograms (kg)) of lot l in market m on day t. $I_{m,l}$ is an indicator variable that is equal to one if lot l is traded in the treatment market, and zero otherwise. $M_{w,l}$ is an indicator variable that is equal to one if lot l is traded w weeks prior to the implementation, and zero otherwise. $Q_{m,t,l}$ is the size (in 100 kg) of lot l sold in market m on day t. $N_{m,t}$ is the total number of lots arriving in market m on day t. $S_{m,t}$ is the total supply arriving in market m on day t. $R_{m,t}$ is the total number of traders in market m on day t. $B_{m,t,l}$ is the number of bids received by lot l in market m on day t. Given this specification, the parallel trends assumption is satisfied if the estimates of γ_w are not statistically significant. The estimation results (Table O.3 in the online appendix) confirm that four out of the five coefficients are not statistically significant, providing support for the parallel trends assumption (Cui et al. 2019, Ganesh et al. 2019).

6.4. DID Model and Results

We estimate the following model as our main DID specification:

$$\log(P_{m,t,l}) = \gamma I_{m,t,l} + \delta_1 \log(Q_{m,t,l}) + \delta_2 \log(N_{m,t}) + \delta_3 \log(S_{m,t}) + \delta_4 R_{m,t} + \delta_5 B_{m,t,l} + (\text{market, date fixed effects}) + \epsilon_{m,t,l}. \tag{4}$$

The variable $I_{m,t,l}$ is the treatment dummy: $I_{m,t,l} = 1$ for all lots in the treatment market on all dates after the implementation, and $I_{m,t,l} = 0$ otherwise. The remaining variables are defined the same as in Equation (3). We control for market and date fixed effects. Standard errors are clustered at the market and date level to account for potential correlation across observations (Bertrand et al. 2004). The coefficient of interest is γ. A positive and significant value of γ indicates that the implementation of the two-stage auction has led to a significant price increase in the treatment market.

In addition, we also estimate the following model to examine whether the treatment effect changes over time as the market participants gain experience with the new auction design:

$$\log(P_{m,t,l}) = I_{m,t,l} \times \left(\sum_{t'=1}^{4} \gamma_{t'} M_{t',l}\right) + \delta_1 \log(Q_{m,t,l}) + \delta_2 \log(N_{m,t}) + \delta_3 \log(S_{m,t}) + \delta_4 R_{m,t} + \delta_5 B_{m,t,l} + (\text{market, date fixed effects}) + \epsilon_{m,t,l}. \tag{5}$$

The variable $M_{t',l}$ is an indicator variable that is equal to one if lot l is traded in the t'th month after implementation, and zero otherwise. All remaining variables follow from Equation (4).

Table 3 presents the estimation results of Equations (4) and (5). We observe from column (2) that the implementation of the two-stage auction has yielded a statistically significant 3.6% increase in tur prices in the

Table 3. Estimated Impact of Implementing the Two-Stage Auction on Tur Prices

Variable	Equation (4)	Equation (5)
$I_{m,t,l}$	0.036*** (0.007)	—
$I_{m,t,l} \times M_{1,l}$ (February)	—	0.015* (0.008)
$I_{m,t,l} \times M_{2,l}$ (March)	—	0.043*** (0.006)
$I_{m,t,l} \times M_{3,l}$ (April)	—	0.031*** (0.009)
$I_{m,t,l} \times M_{4,l}$ (May)	—	0.053*** (0.014)
Other Controls	Y	Y
Observations	30,867	30,867
R^2	0.305	0.306

Notes. "—" means the variable is not present in the model. Standard errors (in parentheses) are clustered at the market and date level. Other controls include $\log(Q_{m,t,l})$, $\log(N_{m,t})$, $\log(S_{m,t})$, $R_{m,t}$, $B_{m,t,l}$, and market and date fixed effects. We omit their coefficient estimates for brevity. See Online Appendix O.3 for detailed results.

$^*p < 0.1$; $^{***}p < 0.01$.

treatment market. In addition, column (3) shows that the positive impact of the implementation is consistent over time (except for a smaller impact in February, potentially due to an initial learning phase). The estimated 3.6% average price increase translates into a revenue gain of $346,000 (USD) for over 10,000 tur farmers who traded in the treatment market in a matter of three months—that is, an 11% increase in their monthly income.[16] Given low profit margins for these farmers (4%–7%), the range of profit improvement is substantial (55%–94%). These positive results are particularly significant, given that the launch of the UMP itself had not significantly improved tur prices prior to the implementation of the two-stage auction.

We perform a number of robustness analyses to further strengthen our results. In particular, we consider (i) a placebo test; (ii) removing control variables; (iii) adding market-level time trends; (iv) filtering data on the subset of traders who were active both preimplementation and postimplementation; (v) filtering data to remove potential time-related confounds; (vi) including data from the 2020 season; (vii) analyzing the impact of the implementation on all bids; (viii) alternative clustering of standard errors; and (ix) aggregating the data at the day level. Online Appendix O.4 presents the details of these analyses. The statistically significant, positive impact of the implementation discussed here survives all of these robustness tests.

6.5. Distributional Impact

Quality is an important factor that determines prices for agricultural products. Thus, it is of great interest to examine whether the impact of the implementation on tur prices depends on a lot's quality. Because the quality of a lot is not directly observable, we use quantile regression analysis based on the main DID model (Equation (4)) to examine the impact of the implementation on the 10th, 25th, 50th, 75th, and 90th percentiles of the price distribution. The coefficient estimates of the treatment dummy are presented in Table 4.

The results demonstrate that the implementation has resulted in a positive impact across all five percentiles of the distribution. This observation confirms the overall positive impact of the implementation on tur prices shown previously. In addition, we observe that the positive impact is stronger at the tails of the distribution, suggesting differential price impacts of the implementation depending on product quality. We learned from our field interactions that traders who mainly purchase the highest-quality tur are distinct from those who mainly purchase the lowest-quality tur, largely due to differences of their buyers. Specifically, the former group primarily sells to buyers who sell the tur in the retail market for direct consumption. In contrast, the latter group sells to processors who mix the tur with other materials to make processed products. The stronger impact of the implementation at the tails of the price distribution suggests that the new auction may have motivated these two groups of traders to exert more focused bidding on the lots that match their desired quality level, hence intensifying competition within each group.

6.6. Postimplementation Insights on the Traders' Bidding Behavior

The empirical analysis thus far demonstrates that the field implementation has led to a significant revenue benefit for farmers in the treatment market. We next use the auction and interview data to provide suggestive evidence of the behavioral insights previously learned from the trader interviews before the implementation. We remark that the purpose of the subsequent analysis is not to establish anticipated regret as the only mechanism driving the traders' bidding behavior in the two-stage auction. The noncontrolled field setting and small sample of traders prevent us from providing a general causal proof. Instead, our analysis is observational in nature and intends to shed light on the role that anticipated regret might have played in affecting the traders' bidding strategies. A causal analysis of the role of anticipated regret in two-stage auctions is a valuable direction for future research.

6.6.1. Anticipated Regret. To shed light on the role of anticipated regret, we use the traders' stated intensity of regret felt in the case of nonqualification in the interviews (Section 4.2.1) to divide the traders in the treatment market into two groups. The traders who do not exhibit any regret in the interviews (stated score for regret = 1) are defined as the "low-regret" group, and

Table 4. Estimated Distributional Impact of the Implementation: Quantile Regression Results

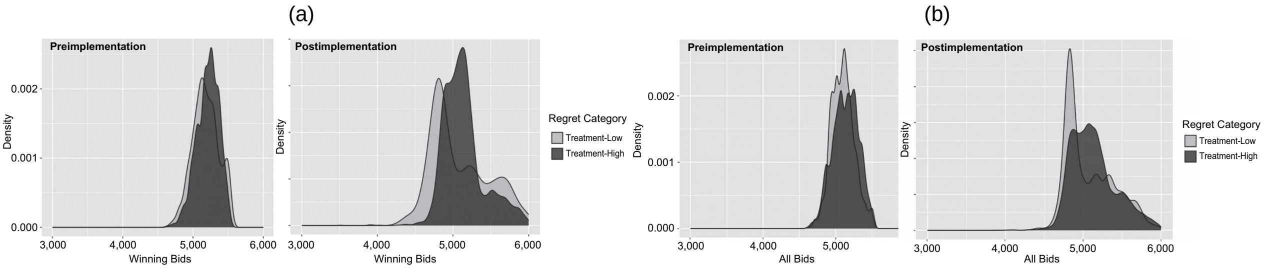
Variable	10th percentile	25th percentile	50th percentile	75th percentile	90th percentile
$I_{m,t,l}$	0.036***	0.018**	0.008***	0.012***	0.026***
	(0.003)	(0.002)	(0.001)	(0.002)	(0.002)
Other controls	Y	Y	Y	Y	Y
Observations	30,867	30,867	30,867	30,867	30,867

Notes. Bootstrapped standard errors are included in the parentheses. Other controls include $\log(Q_{m,t,l})$, $\log(N_{m,t})$, $\log(S_{m,t})$, $R_{m,t}$, $B_{m,t,l}$, and market and date fixed effects. We omit their coefficient estimates for brevity.
$^{**}p < 0.05$; $^{***}p < 0.01$.

those who show regret (stated score for regret ≥ 5) are defined as the "high-regret" group. Figure 5(a) (Figure 5(b)) illustrates the distribution of the winning bids (all bids) by traders in the high-regret (dark gray) and low-regret (light gray) groups preimplementation (left) and postimplementation (right). Both figures suggest that the difference in the bid distribution between the high-regret and low-regret traders becomes more pronounced postimplementation. Results from t-tests confirm that postimplementation, the winning bids ($t = 2.15$, $p = 0.03$) and all bids ($t = 12.78$, $p < 0.01$) by high-regret traders are significantly higher than those by low-regret traders, consistent with the results of our behavioral auction model. We were not able to conduct interviews with traders in the control market due to logistical and time constraints. Given this data limitation, a triple DID specification cannot be estimated. Nevertheless, we conduct additional regression analysis to further evaluate the above observations in Online Appendix O.5.

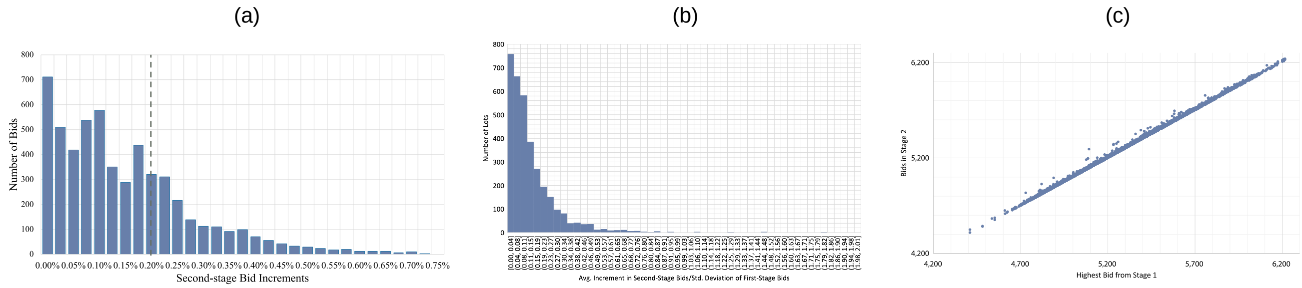
6.6.2. Second-Stage Bidding. We analyze the auction data postimplementation to demonstrate that the traders' second-stage bidding behavior is indeed consistent with the qualitative findings in Section 4.2.2, as well as the model predictions in Section 5. First, Figure 6(a) presents the distribution of the relative bid increments in the second stage among all lots that received second-stage bids. The relative bid increment for a lot is calculated as a qualified trader's second-stage bid for the lot minus the highest first-stage bid, divided by the highest first-stage bid. We observe that the relative bid increment in the second stage never exceeds 0.75%, with an average of 0.2% (9 rupees/100 kg compared with the average price of about 5,000 rupees/100 kg). Second, Figure 6(b) presents the distribution of the ratio between the average increment in the second-stage bids and the standard deviation of the first-stage bids across all lots that received second-stage bids. We find that the average ratio is 14.8%, and the median is 9.6%. Thus, on average, the average increment in the second-stage bids for a lot is only 14.8% of the variation in the first-stage bids for that lot. Finally, we observe from the data that a total of 2,365 lots (68% of all the lots that received second-stage bids) were won by traders ranked in the second or third place after the first stage. Figure 6(c) presents a scatter plot of *all* second-stage bids against the highest first-stage bid for all the lots that received second-stage bids. It is evident from this figure that all bids submitted in the second stage (including those from the second- and third-ranked traders) are very close to the disclosed highest first-stage bids. In particular, the average bid increment over the highest first-stage bid is 8.86 rupees/100 kg and 8.34 rupees/100 kg among second- and third-ranked traders, and the difference is not statistically significant (t test, $t = 1.48$, $p > 0.1$).[17] Taken together, these results provide evidence of limited bid increase in the second stage of the two-stage

Figure 5. Comparing Bids from High-Regret and Low-Regret Traders in the Treatment Market Preimplementation and Postimplementation



Notes. (a) Distribution of winning bids. Panel (a) presents the distribution of winning bids at the lot level in the treatment market preimplementation (left) and postimplementation (right) among high-regret (dark gray) and low-regret (light gray) traders. (b) Distribution of all bids. Panel (b) presents the distribution of all bids at the lot level in the treatment market preimplementation (left) and postimplementation (right) among high-regret (dark gray) and low-regret (light gray) traders.

Figure 6. (Color online) Traders' Bidding Behaviors in the Second Stage of the Two-Stage Auction Postimplementation



Notes. (a) Distribution of relative bid increments in the second stage. (b) Distribution of the ratio between the average increment in second-stage bids and the standard deviation of first-stage bids. (c) Second-stage bids vs. highest first-stage bid. The relative bid increment for a lot in (a) is calculated as a qualified trader's second-stage bid minus the highest first-stage bid, divided by the highest first-stage bid. The vertical line indicates the average second-stage bid increment across all lots and traders. The average increment in second-stage bids for a lot in (b) is calculated as the average difference between the second-stage bids and the highest first-stage bid for that lot. The figures only account for those lots that indeed received second-stage bids.

auction, consistent with our qualitative findings in the trader interviews (Section 4.2.2) and predictions from the behavioral auction model (Lemmas 1 and 2).

7. Conclusions

This paper introduces a behavior-centric and field-based methodology to design effective auction mechanisms to enhance farmers' revenue in online agri-platforms. By collaborating closely with the state government of Karnataka, we *design*, *analyze*, and *implement* a new two-stage auction on the state's agri-platform for a major lentils market. A difference-in-differences analysis based on lot-level auction data demonstrates significant revenue gain for over 10,000 smallholder farmers traded in the market in a matter of three months. Encouraged by the positive outcomes and guided by the research insights, the state government has selected a set of suitable commodities and markets to next implement the two-stage auction on a larger scale. The methods introduced in this paper provide generally applicable knowledge to researchers and platform designers as they continue to enhance the design of agri-platforms to improve the livelihood of smallholder farmers.

This work also opens a number of avenues for future research. First, a causal analysis of the role that anticipated regret plays in two-stage auctions (both in laboratory and field settings) could be valuable for further refining the design features of two-stage auctions. Second, investigating the influence of other behavioral factors, such as overconfidence (Allen and Evans 2005) and gamification cognitive biases (Hamari et al. 2015), which may become salient in two-stage auctions, can contribute to further enhancements in the auction design. Finally, although our focus has been on farmers' revenue, examining the impact of these interventions on downstream outcomes can be equally important for policymakers in developing countries.

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Endnotes

[1] Specifically, bidders are assumed to be risk-neutral, only concerned about monetary payoffs, and perform Bayesian equilibrium reasoning.

[2] If a lot receives fewer than k bids in the first stage or if no one increases their bids in the second stage, then the top bidder in the first stage is declared as the winner of the lot.

[3] In the interviews, we were careful not to collect any identifiable information that can trace back to individual traders, and we also assured the traders of the confidentiality of their responses. In particular, before the data were shared with us for the analysis, our collaborator anonymized the data to remove any identifiable information. Each trader was assigned a numeric ID in the data. We did not provide monetary compensation for the traders; nevertheless, each trader who participated in the interview was presented at the end with a small gift (a box of cookies) as a token of appreciation.

[4] Athey et al. (2011) use a similar setup in their study of timber auctions.

[5] Within this hour, a considerable portion of the time was spent to carefully understand the trading processes and market dynamics. Furthermore, all of the interactions required translation between English and the local language to ensure that the traders correctly understood the questions and the researcher recorded the answers properly, all of which were very time-consuming.

[6] The type of regret discussed here corresponds to loser's regret. Scholars have also studied the effect of winner's regret in auctions. Winner's regret may arise if the winning bidder can observe the second highest bid at the end of the auction and realize that they could have won with a lower bid. However, prior research shows that winner's regret has a limited effect in single-stage auctions (Filiz-Ozbay and Ozbay 2007). In our context, the second highest bid is never disclosed to the traders in either the original auction or the two-stage auction. Thus, we do not expect winner's regret to be salient in our setting. Alternatively, winner's regret may be relevant if the traders actively compare the winning bids of similar lots after the auctions are completed, either within the same day or across days. We postulate that this is unlikely for the following reasons.

First, these are high-volume markets with hundreds of lots being sold in a day, and winner declaration for all lots occurs at the same time. Second, we learned from our interactions with the traders that after the winner declaration, the traders are largely occupied by laborious and time-consuming postauction logistics, including weighing, sorting, aggregating the lots from multiple commission agents, arranging transportation, and transferring payments to the farmers. The traders must complete these tasks by the end of the day. Given these reasons, the traders are unlikely to have time to compare winning bids, and, hence, winner's regret would play a limited role.

[7] We initially designed out-of-context questions to elicit the traders' behavioral factors following the literature. However, our collaborator commented that the quality of responses to these questions would be low because the traders were very busy and would be uninterested in answering out-of-context questions unrelated to their businesses. Indeed, when we pilot-tested the out-of-context questions with some traders, they showed an apparent lack of interest and gave us random answers. Therefore, we ultimately decided to utilize questions related to the bidding context to evaluate the behavioral factors.

[8] One of the traders interviewed only had time to answer our questions regarding the trading process in the market and did not complete the later questions in the survey. As a result, we do not have his response to the question shown in Figure 3.

[9] To complement the trader interviews, we conducted an out-of-sample online study to show that the two-stage auction would *intensify* bidders' anticipated regret, as compared with the single-stage first-price auction. We refer the reader to Online Appendix O.1 for more details about this online study.

[10] We assume that ties are broken randomly in the auction. Because we consider a continuous distribution for the traders' valuations, ties occur with zero probability, and, hence, this tie-breaking assumption does not affect the traders' expected payoff function.

[11] We drop the small increment ϵ in the expected payoff here and wherever applicable subsequently for expositional simplicity. This is justified for practical purposes because ϵ is orders of magnitude smaller than the bid price (also see Section 6.6.2). Including ϵ will not change the model insights, but will overcomplicate the mathematical expressions.

[12] There was a gap of multiple months between the interviews and the implementation. Given this intentional gap and the busy schedule of the traders, it is unlikely that the interview interactions would have a spillover effect on the traders' bidding behaviors after the implementation.

[13] Because the second-stage bidding is done within a short time, we do not expect increased risk of collusion from the two-stage auction.

[14] Because our analysis is based on major markets, the resulting insights may not hold for smaller markets.

[15] We use this cutoff date because market officials confirm that the 2019 season for tur ends by the last week of May in the state. Aggregate supply data from the UMP shows that approximately 93% of the total quantity traded in 2019 was before June.

[16] We use the total number of lots sold in the treatment market as a proxy for the total number of farmers who traded in the market. Because the same farmer may visit and sell in the market for multiple days, the stated 11% income raise is a conservative estimate.

[17] Elmaghraby et al. (2012) study the impact of rank feedback versus full-price feedback on bidders' bidding strategies in laboratory experiments. They find that rank feedback results in lower average prices than full-price feedback in procurement auctions. In our case, rank feedback appears to have a limited effect on the traders' second-stage bidding, as compared with the effect of the highest first-stage bid. This can be seen in the data in two aspects. First, the second-stage bids are very close to the highest first-stage bid (Figure 6(a)). Second, *both* second- and third-ranked traders behave very similarly in the second stage; that is, they try to outbid the highest first-stage bid slightly, regardless of their rank (Figure 6(c)).

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